

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Component	Short Description
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU (Processor)	The "brain" of the computer that performs calculations and executes instructions.
CPU Cooler (Heatsink & Fan)	Keeps the processor cool by removing excess heat generated during operation.
RAM (Memory)	Temporary storage that holds data and programs currently being used by the CPU.
Storage Drive (SSD/HDD)	Stores the operating system, software, and user files permanently.
Power Supply Unit (PSU)	Converts AC power from the wall into DC power for computer components.
Graphics Card (GPU)	Processes graphics and video output, especially important for gaming and video editing.
Case Fans	Improve airflow inside the cabinet to keep components cool.
Optical Drive (Optional)	Reads and writes CDs, DVDs, or Blu-ray discs. Not found in most modern PCs.
Expansion Cards	Additional cards such as Wi-Fi, sound, or network cards that add extra features.
Cables and Connectors	Connect power and data between components such as the motherboard, drives, and PSU.
CMOS Battery	Small battery that maintains BIOS settings and system clock when the PC is turned off.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

✓ Safety Precaution / Tool	Why It Is Important
☐ Turn Off the Computer	Prevents electrical shock and protects components from damage.
☐ Unplug the Power Cable	Ensures no electricity is flowing into the system while working.
☐ Disconnect All External Devices	Prevents accidental damage to connected peripherals and cables.
☐ Use an Anti-Static Wrist Strap	Protects sensitive electronic components from static electricity (ESD).

✓ **Safety Precaution / Tool**

Why It Is Important

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| ☐ Work on a Clean, Flat Surface | Reduces the risk of losing parts and prevents accidental damage. |
| ☐ Avoid Carpeted Areas | Minimizes the buildup of static electricity. |
| ☐ Keep Hands Dry | Prevents electrical hazards and protects components. |
| ☐ Handle Components by the Edges | Avoids damage to circuits, chips, and connectors. |
| ☐ Store Screws in a Small Container | Prevents screws from getting lost during the process. |
| ☐ Read the Hardware Manual | Helps identify components and avoid incorrect installation or removal. |